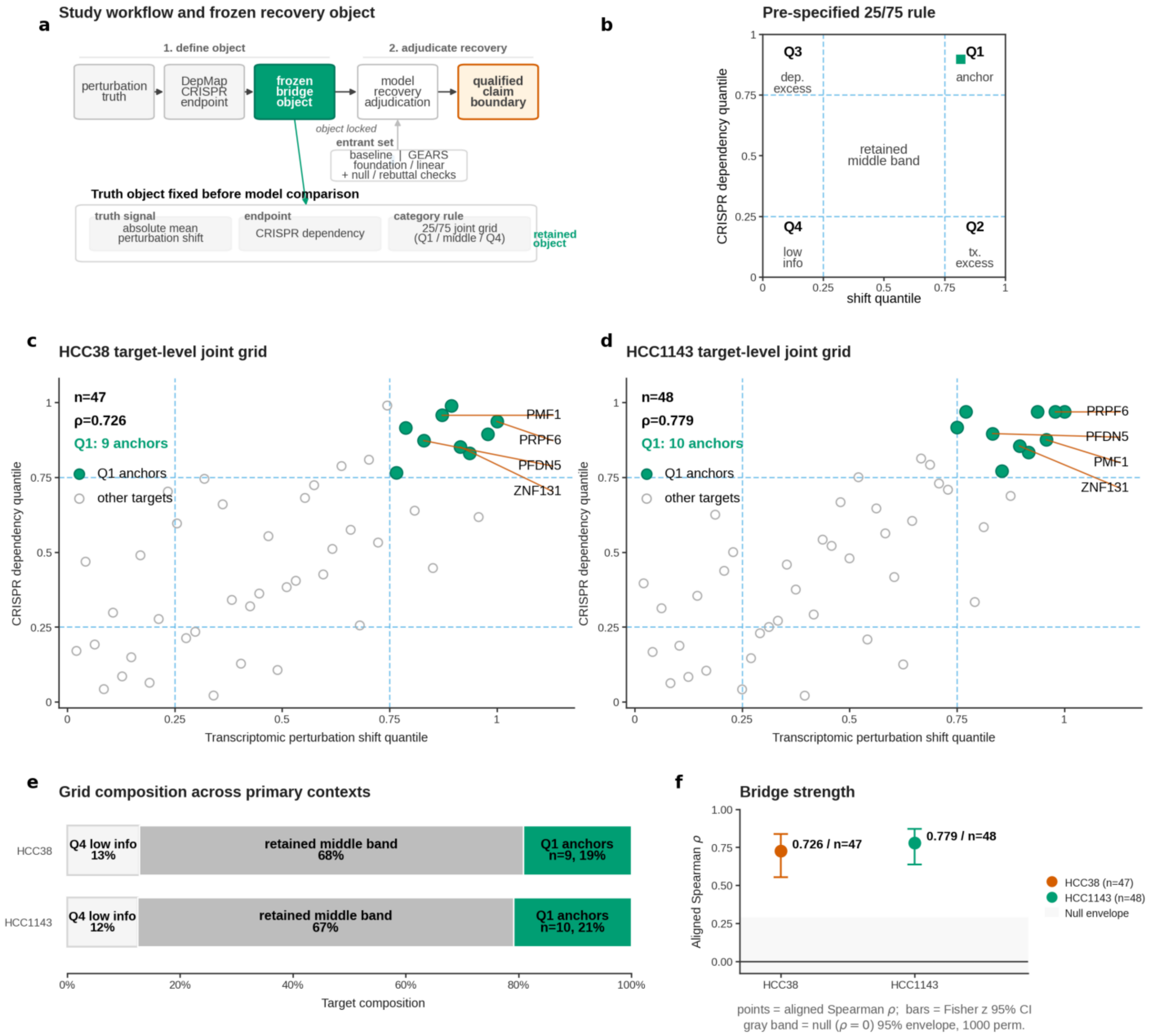


Fig. 1 | Truth object definition. a, Workflow defining the frozen phenotype-aligned recovery object before model scoring by linking absolute mean perturbation shift to CRISPR DepMap dependency. b, Pre-specified 25/75 joint-percentile rule separating Q1 anchors, Q2 transcriptomic excess, Q3 dependency excess, Q4 low-information targets and the retained middle band. c,d, HCC38 and HCC1143 target-level rank-percentile grids; panels report n, aligned Spearman rho and Q1 anchor count. e, Category composition across the two primary contexts, including zero-count Q2 and Q3 categories. f, Bridge strength summary showing observed aligned Spearman rho with Fisher z 95% confidence intervals and a 1,000-permutation null envelope.

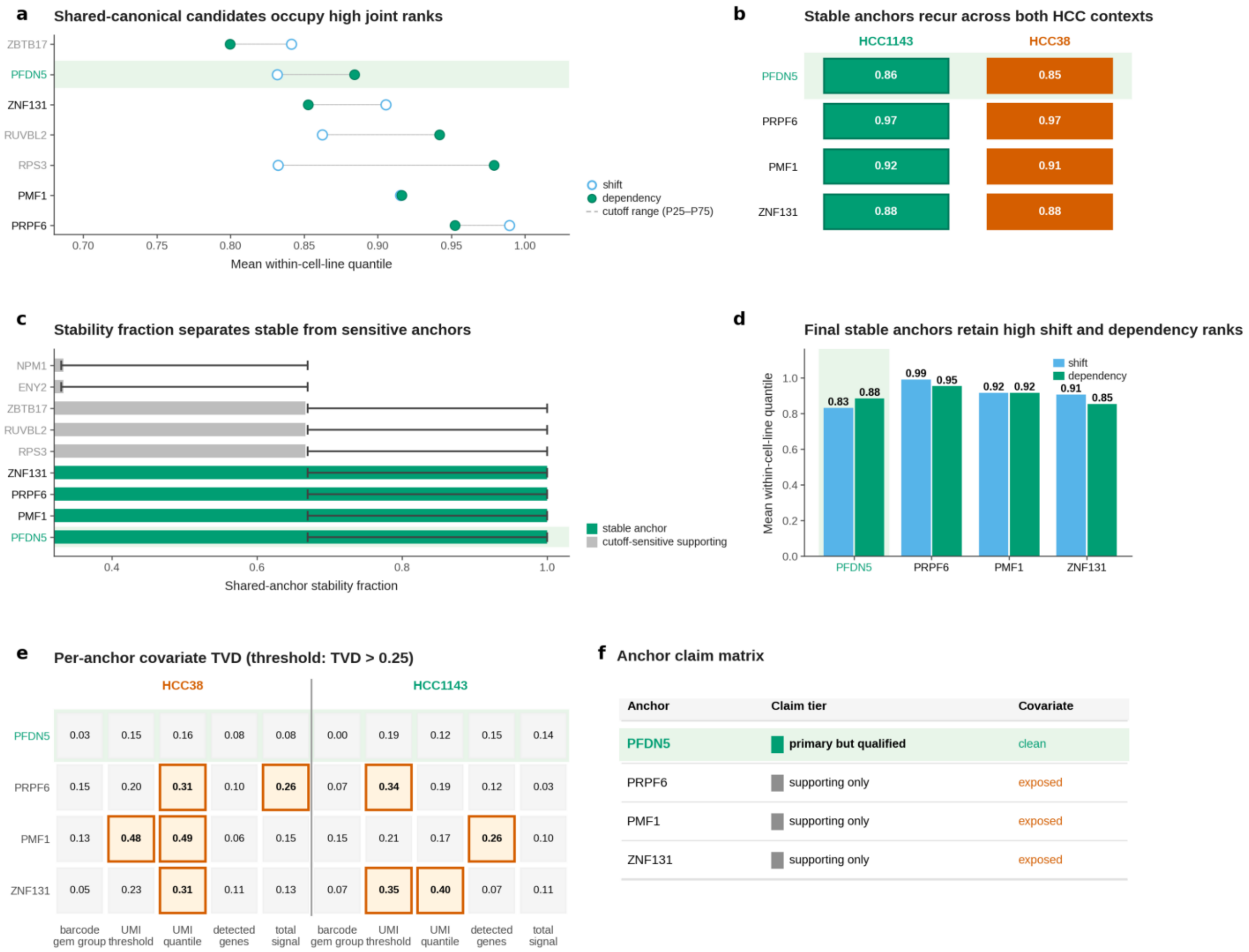


Fig. 2 | Anchor tiering. a, Shared-canonical anchor ranking by paired shift and dependency quantile means across HCC38 and HCC1143. b, Recurrence matrix for the four final stable anchors, with Q1 status and mean joint quantiles shown per context. c, Stability fraction separates stable anchors from cutoff-sensitive supporting objects without using covariate information. d, Stable anchors retain high shift and dependency ranks across both HCC contexts. e, Per-anchor covariate Total Variation Distance (TVD) audit across five covariate axes and two HCC contexts; TVD > 0.25 marks imbalance. f, Claim-tier matrix: PFDN5 is primary but qualified, whereas PMF1, PRPF6 and ZNF131 remain supporting because of covariate exposure.

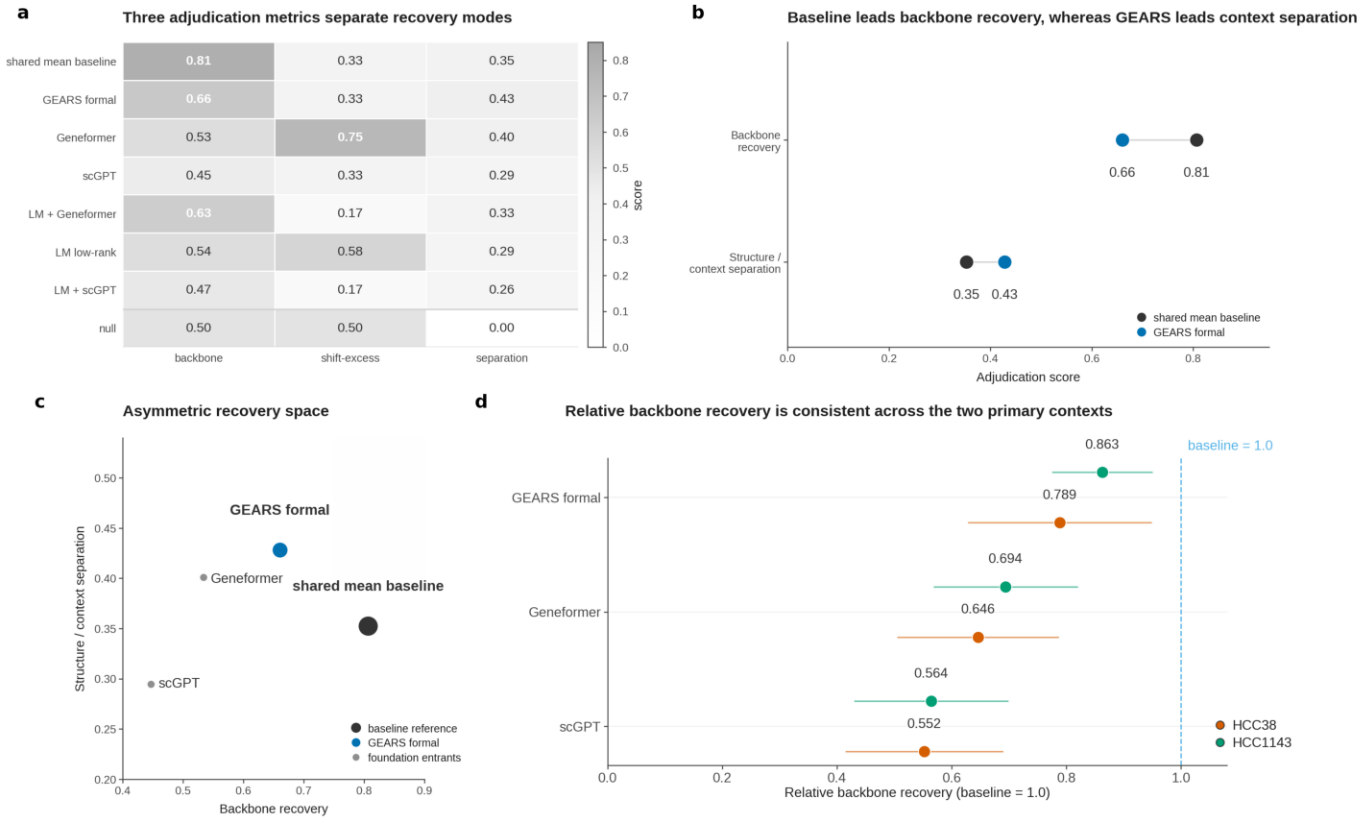
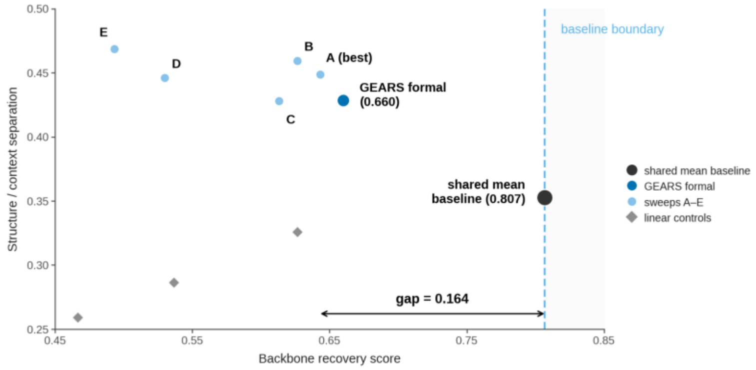


Fig. 3 | Model adjudication. a, Three pre-specified metrics evaluate backbone recovery, shift-excess identification and structure-versus-context separation across the shared-mean baseline, GEARS, foundation-model entrants, linear controls and a null reference. b, Paired summary showing that the shared-mean baseline leads backbone recovery, whereas GEARS leads structure-versus-context separation. c, Backbone recovery versus structure-versus-context separation places entrants in an asymmetric recovery space; no entrant occupies the illustrative upper-right region, so GEARS is retained as an architecture-level diagnosis rather than an overall primary winner. d, Per-context backbone recovery ratios relative to the shared-mean baseline are shown for GEARS, Geneformer and scGPT in HCC38 and HCC1143, confirming that the residual backbone gap is present in both primary contexts.

a Pre-specified finite-budget GEARS candidates

Candidate	Epoch	LR	WD	Role
GEARS formal	30	1×10^{-3}	1×10^{-6}	reference recipe
A (best)	30	2×10^{-3}	1×10^{-6}	best sweep candidate
B	20	1×10^{-3}	1×10^{-6}	sweep candidate
C	30	1×10^{-3}	1×10^{-5}	sweep candidate
D	30	5×10^{-4}	1×10^{-6}	sweep candidate
E	40	1×10^{-3}	1×10^{-6}	sweep candidate

b Prespecified rebuttal candidates do not close the backbone gap



c No tested rebuttal candidate closes the backbone gap to the shared-mean baseline

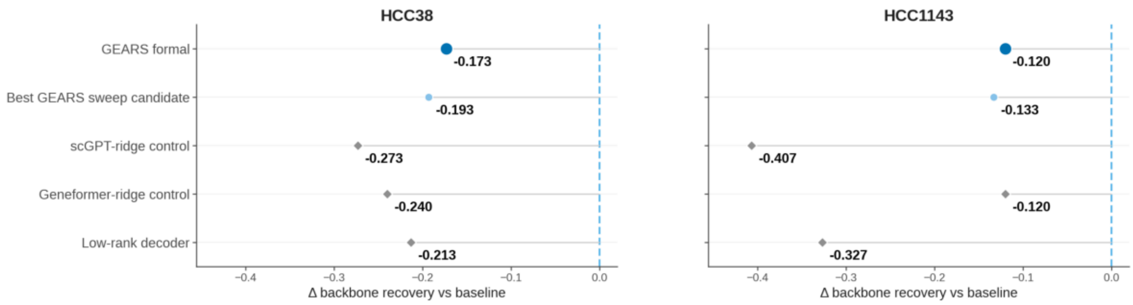


Fig. 4 | Finite-budget rebuttal tests. a, Six pre-specified GEARS recipes define the finite local rebuttal neighborhood under the unchanged truth object and scoring system. b, Rebuttal recovery space compares the shared-mean baseline, formal GEARS, five GEARS sweep candidates and three embedding-based linear controls on backbone recovery and structure/context separation. c, Per-context residual backbone gaps show that all tested candidates remain below the shared-mean baseline in HCC38 and HCC1143; the stop rule therefore does not promote GEARS to the primary winner.

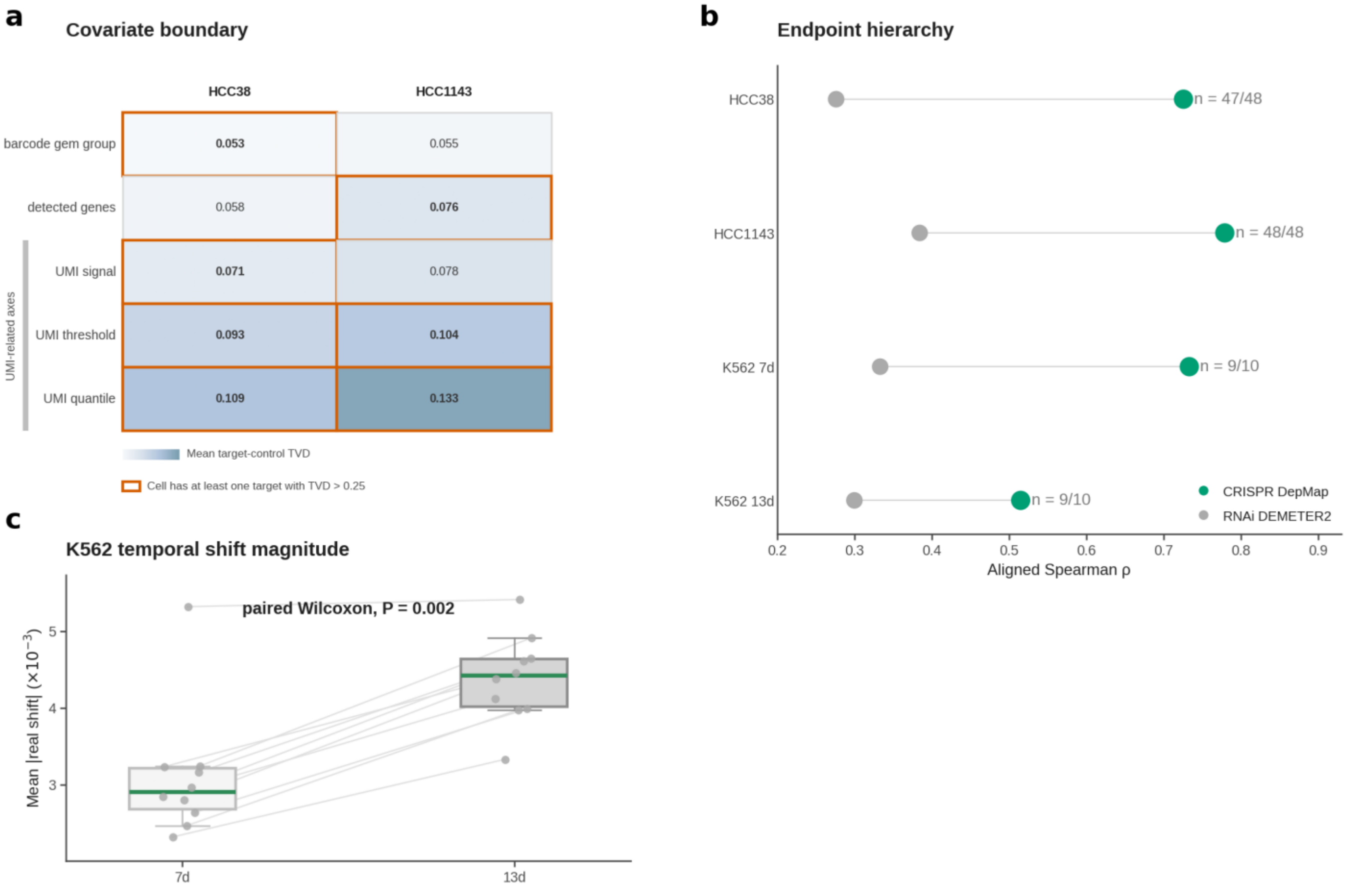


Fig. 5 | Boundary governance. a, Covariate boundary in HCC38 and HCC1143, showing mean target-control TVD across five covariate stratifications and marking targets exceeding TVD > 0.25. b, Endpoint hierarchy across HCC38, HCC1143, K562 7d and K562 13d compares CRISPR DepMap dependency with RNAi DEMETER2; CRISPR is higher in all four contexts. c, K562 temporal stratification shows larger perturbation magnitude at 13d but stronger dependency-aligned rank structure at 7d. d, A0/A1/B tiering retains K562 as architecture-form and bounded bridge-form support, not content-level replication.